

Virtual Reality for Remote Controlled Robotics in Engineering Education

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ABSTRACT

There is a high demand for high-end lab equipment in engineering education, especially for courses that require practical hands-on lab exercises. However, this equipment is quite expensive which forces some institutions to seek other alternatives or forego them altogether. In addition, online courses tend to be light on hardware because of the practical and technical challenges associated with running remote hardware-based laboratories. Therefore, a system is needed that allows students who are enrolled in remote courses to experience hardware-intensive classes just as they would in an in-person course (especially in light of the COVID-19 pandemic). In this work, we use virtual and augmented reality to build and test a remote UR-10 robotics lab that allows students to work together on a hands-on robotics-based lab. In the future, we plan to demonstrate its technical and pedagogical suitability by running a user study in a real-world class whereby remote students enrolled in a course will participate in a series of experiments to control a UR-10 robot arm. It opens up avenues to explore the advantages and shortcomings of running coordinated group lab exercises between students from different institutions, time-zones, and continents by time-decoupling the schedules. The use of a Virtual Reality interface coupled with a digital twin of the UR10 robot will allow us to investigate the suitability of different user interfaces, and what VR tools would best facilitate remote group work and faculty instruction.

Index Terms: Robots, VR, AR, Education, Remote Laboratories

1 INTRODUCTION

The challenges facing the teaching of engineering courses largely stem from the diverse academic backgrounds of students and instructors in addition to the limited availability, or the lack thereof, of high-end equipment needed to put theoretical concepts into practice. For most engineering courses to be effective, a significant portion of the syllabus/curriculum must include laboratory or practical hands-on exercises. World-class universities and universities with funding usually have these hardware components available for their students. However, they remain unused during the vast majority of the day, for example during the night. In addition, students have different aptitude/speeds of learning, and though some students would like to go over and above the required coursework, they might not get the opportunity to. This is not only because the lab equipment free-time is not available - the obvious reason - but also, because the expensive equipment requires supervision and moderation - a role usually played by lab-techs, teaching assistants and faculty. Conversely, institutions of higher learning in developing countries lack this equipment, making the learning in some of the courses less effective as the students miss out on critical laboratory components needed for optimum internalization of the concepts taught in class.

Online and remote education has gone some way in alleviating a fraction of these challenges, particularly by offering online class-

room management tools thus allowing access to world class faculty by remote learners in any part of the world by the click of a button [1, 2]. However, much remains and needs to be done to make the teaching of online courses with significant lab components. Some Massively Open Online Courses (MOOCs) overcome this by using relatively affordable and standardized electronic kits which the students purchase and use for their experiments. In this demo, we showcase a robotics lab exercise that can be done collaboratively by groups of students from remote locations. In doing so, we include several levels of access by students (tablet, desktop and VR) to make the system more accessible. In addition, we include several tools, like video and voice, to facilitate communication and monitoring of the lab experiment. We also include tools that allow the moderation of a classroom by the person in charge.

2 DEMO SET-UP

We implemented the VR and AR components of our system in Unity3D (version 2019.4.1f1). A dedicated laptop running a python server was connected to the UR-10 robot using an Ethernet cable. Satellite tablets and computers will access the robot through a network relay which allows us to circumvent the university's stringent firewall. Programming of the robot was done using a custom scripting python-like language (UR-script) and this script was uploaded to the robot at run-time, allowing information exchange and remote control of the robot. For low latency response of the robot, the uploaded script runs a loop that checks the values of six registers for the robot's next pose, and a seventh one to open or close the Robotiq 2-finger adaptive gripper. These values are updated by the python server based upon the information received from the user application.

To run the demo, the user needs to download the app to their system (VR, tablet or computer and load it up). When run, the app allows them to connect to the robot if an instructor has a valid session up and running. Once in the class environment, the user can perform a lab task with the robot. A virtual robot, running on top of a UR-10 simulator, allows the user to try different motion profiles until they're happy with the accuracy and the trajectory of the robot arm. Following this, the user can send the command to move the real robot arm.

The particular lab task is a simple pick and place task that involves the stacking of pallets in the middle of a play area. This is a common use of robots in an industrial setting.

Figure 2 below shows an overview of the set-up. The left hand side shows the user interface. The user may run the application on a tablet, a PC or on a VR HMD. The 3D model of the virtual robot (the red/yellow image below) can be moved as many times as the user wishes until the user is happy. This runs on top of a UR-10 simulator which generates a motion profile that is identical to that of the real robot controller. The two digital twins of the simulator and the physical robot serve as visual feedback to the user. An Azure Kinect brings a video of the actual robot to the user tablets.

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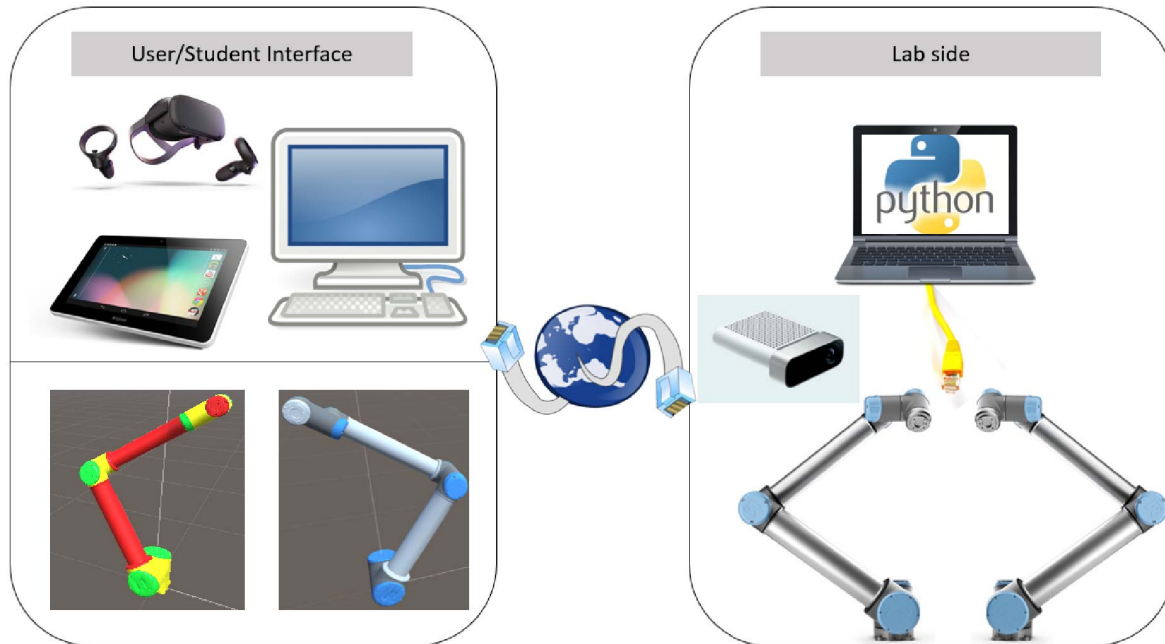


Figure 1: Overview of the set-up. The user has the option of remotely controlling the robot using a tablet, a computer or a virtual reality headset. A python server servers as the bridge between the remote commands and the UR-10 robot arm (right)

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